

Original Investigation

Evaluating Prediction Models of Sleep Apnea From Smartphone-Recorded Sleep Breathing Sounds

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Key Points

Question Can smartphone-recorded sound analysis help predict obstructive sleep apnea?

Findings In this cross-sectional study of 423 patients who visited a sleep center for snoring or sleep apnea, sleep breathing sounds were recorded using a smartphone during routine, full-night, in-laboratory polysomnography. Prediction models of obstructive sleep apnea through sleep breathing sound produced accuracies of 88.2%, 82.3%, and 81.7% for an apnea hypopnea index threshold of 5, 15, and 30 events/h, respectively.

Meaning The findings of this study suggest that smartphone-recorded sleep breathing sounds may be used to create prediction models for obstructive sleep apnea, but additional verification of real-life recordings using various smartphone devices should be conducted.

Abstract

Importance Breathing sounds during sleep are an important characteristic feature of obstructive sleep apnea (OSA) and have been regarded as a potential biomarker. Breathing sounds during sleep can be easily recorded using a microphone, which is found in most smartphone devices. Therefore, it may be easy to implement an evaluation tool for prescreening purposes.

Objective To evaluate OSA prediction models using smartphone-recorded sounds and identify optimal settings with regard to noise processing and sound feature selection.

Design, Setting, and Participants A cross-sectional study was performed among patients who visited the sleep center of Seoul National University Bundang Hospital for snoring or sleep apnea from August 2015 to August 2019. Audio recordings during sleep were performed using a smartphone during routine, full-night, in-laboratory polysomnography. Using a random forest algorithm, binary classifications were separately conducted for 3 different threshold criteria according to an apnea hypopnea index (AHI) threshold of 5, 15, or 30 events/h. Four regression models were created according to noise reduction and feature selection from the input sound to predict actual AHI: (1) noise reduction without feature selection, (2) noise reduction with feature selection, (3) neither noise reduction nor feature selection, and (4) feature selection without noise reduction. Clinical and polysomnographic parameters that may have been associated with errors were assessed. Data were analyzed from September 2019 to September 2020.

Main Outcomes and Measures Accuracy of OSA prediction models.

Results A total of 423 patients (mean [SD] age, 48.1 [12.8] years; 356 [84.1%] male) were analyzed. Data were split into training (n = 256 [60.5%]) and test data sets (n = 167 [39.5%]). Accuracies were 88.2%, 82.3%, and 81.7%, and the areas under curve were 0.90, 0.89, and 0.90 for an AHI threshold of 5, 15, and 30 events/h, respectively. In the regression analysis, using recorded sounds that had not been denoised and had only selected attributes resulted in the highest correlation coefficient ($r = 0.78$; 95% CI, 0.69-0.88). The AHI ($\beta = 0.33$; 95% CI, 0.24-0.42) and sleep efficiency ($\beta = -0.20$; 95% CI, -0.35 to -0.05) were found to be associated with estimation error.

Conclusions and Relevance In this cross-sectional study, recorded sleep breathing sounds using a smartphone were used to create reasonably accurate OSA prediction models. Future research should focus on real-life recordings using various smartphone devices.

Introduction

Obstructive sleep apnea (OSA) is a common disorder that is widely known to be associated with decreased quality of life and increased incidence of neurovascular and cardiovascular diseases.¹ The gold standard diagnostic method for OSA is attended, full-night, in-laboratory polysomnography (PSG) that involves recording numerous physiologic signals that are manually scored by certified sleep technicians or physicians. Therefore, in-laboratory PSG is expensive, and accessibility to a sleep facility is not always easy. Considering the high prevalence of OSA, performing full-night in-laboratory PSG may not be practical for all patients. Another major drawback is that the sleep environment during PSG is not exactly the same as in real life, and there is considerable night-to-night variation.²⁻⁴ Consequently, portable home sleep apnea test devices were developed and have been used in selected cases⁵ as

screening tools and for patient monitoring during and after treatment. However, to use these devices, patients must obtain a physician's prescription, and the high cost limits access to the general public.

Meanwhile, sleep breathing sounds, which include snoring, are biomarkers that may represent OSA.^{6,7} In a previous study, we focused on the prediction of OSA from sound data collected during sleep using a conventional camcorder microphone.^{8,9} The advantage of using recorded sleep breathing sounds in the detection of OSA is that it can be repeatedly tested. In addition, the widespread use of personal smartphones, which have a microphone recorder, makes it easy to implement.

The present study focused on creating prediction models for OSA from sleep breathing sounds recorded from a conventional smartphone. In our previous work,^{8,9} after sound was recorded, data were processed, and features were extracted. However, in a smartphone environment, data processing and feature extraction need to be minimized.

The primary goal of this study was to assess whether smartphone-recorded breathing sounds could be used to predict OSA. The secondary goal was to evaluate whether minimizing features and sound processing could influence prediction performance.

Methods

Study Participants and Full-Night PSG

A cross-sectional study was performed. From September 2015 to September 2019, patients who visited the sleep clinic of the Department of Otorhinolaryngology–Head and Neck Surgery at Seoul National University Bundang Hospital because of snoring or sleep apnea were included in this study. All patients underwent an attended, in-laboratory, full-night PSG (Embla N7000 [Medcare Flaga]). Written informed consent was obtained from each participant, and the study complied with the Declaration of Helsinki. This study was approved by the institutional review board of Seoul National University Bundang Hospital.

Apnea was defined as cessation of airflow for at least 10 seconds; hypopnea was defined as a greater than 50% decrease in airflow for at least 10 seconds or a moderate reduction in airflow for at least 10 seconds associated with arousals or oxygen desaturation ($<4\%$).¹⁰ The apnea hypopnea index (AHI) was defined as the total number of apnea and hypopnea events per sleep hour. Patients with analysis time periods (time spent in bed) that lasted at least 4 hours were included.

Sound Recording and Processing

For all patients, audio recordings were carried out using a smartphone (LG G3 [LG Electronics]) during full-night PSG. The smartphone was consistently placed at the corner of a shelf and was approximately 1 m from the patient's head ([Figure 1](#)). Sleep breathing sounds were recorded for all sleep stages, from sleep onset to offset.

Figure 1. Bed and Smartphone Used for Polysomnography



The distance between the patient's head and the smartphone is about 1 m. Sound analyses began with converting audio files into WAV file formats. Audio data were discarded for the initial 30 minutes from initiation of recording, considering the median (IQR) time of sleep latency as 9.0 (11.5) minutes. Analysis was ceased once the analysis time exceeded 6 hours. First, noise filtering was performed with a spectral subtraction using Audicity,¹¹ followed by data segmentation into 5-second windows; thereafter, sound features were extracted from each windowed signal. These procedures were performed using jAudio, a Java-based audio extraction program.¹² We attempted to extract all sound features provided by the software. Means, standard deviations, derivatives of means, and derivatives of standard deviations of each feature were then calculated. Finally, 508 features, representing a variety of temporal and spectral characteristics of the audio signal of sleep breathing sounds, were extracted from each patient (eTable 1 in the [Supplement](#)). The process was repeated for the recorded sounds without noise filtering.

Data Set Portioning and Machine Learning

Partitioning of the data set into training and test data sets and machine learning were performed using the free software WEKA.¹³ The data of the patients were first randomly sorted and divided into training (256 [60.5%] patients) and test (167

[39.5%] patients) data sets by using the Randomize and RemovePercentage filters. Each patient yielded 2 data sets acquired from 2 different conditions (with and without noise filtering).

First, binary classifications were conducted for 3 different threshold criteria of AHI: 5, 15, or 30 events/h. The synthetic minority oversampling technique balancing method (SMOTE) was used to introduce a balance in the training data set.¹⁴ A binary classification of AHI was carried out using a random forest method because this demonstrated the best performance according to a previous study.⁹ A 10-fold cross validation procedure was applied. The final models were then assessed in an independent test data set.

The estimation of AHI was performed by creating regression models, also using the random forest method in the training data set. Regression models were developed based on noise reduction and feature selection. Feature selection involved selection of “n” number of relevant features from the original 508 sound features, and this was completed using the correlation-based Feature Subset Evaluation in WEKA.¹⁵ In total, 4 regression models were developed as follows: noise reduction only (model 1), noise reduction with feature selection (model 2), naive recorded sound only (without noise reduction) (model 3), and naive recorded sound with feature selection (model 4).

Model performance measures, including accuracy, κ index, sensitivity, specificity, F1 score, area under the precision recall curve, and the area under the curve (AUC) of the receiver operating characteristic curve, were computed for the binary classification. The AHI estimation was evaluated; mean absolute error and root mean squared error were used for error metrics. Finally, Shapley Additive Explanation (SHAP) values were calculated to provide attribution values for each feature in the best prediction model by using Python, version 3.5 (Python Software Foundation). The SHAP value evaluates the statistical significance of the output resulting from the inclusion of a particular feature for all other feature combinations.¹⁶

Statistical Analysis

After confirmation of non-normal distribution, a Mann-Whitney U test was used to compare the mean scores of clinical variables of the training and test data sets. A χ^2 test was used for categorical variables. Pearson correlation and multivariate linear regression models were carried out to examine the associations between error and clinical parameters. Age, sex, body mass index (BMI, calculated as weight in kilograms divided by height in meters squared), AHI, and sleep efficiency were used as input variables in the models. Statistical analyses were performed using SPSS, version 22.0 (IBM Corporation). Continuous parametric variables are presented as means (SDs), and 2-sided $P < .05$ was considered statistically significant.

Results

Patient Population

During the study period, 760 patients underwent PSG. After excluding patients younger than 18 years and those who refused to enroll in this study, a total of 423 patients were included. Patients were grouped according to OSA severity: normal ($n = 43$; mean [SD] AHI, 2.2 [1.5] events/h), mild ($n = 80$; mean [SD] AHI, 9.4 [2.8] events/h), moderate ($n = 109$; mean [SD] AHI, 22.0 [4.2] events/h), and severe ($n = 191$; mean [SD] AHI, 55.1 [17.3] events/h). [Table 1](#) summarizes the main characteristics of the analyzed patients. There were no considerable differences in demographic and PSG parameters between the training and test data sets.

Table 1. Clinical Characteristics of Patients

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Clinical characteristics	Mean (SD)			P value
	All (n = 423)	Training data set (n = 256)	Test data set (n = 167)	
Sex, male:female	356:67	214:42	142:25	.63
Age, y	48.1 (12.8)	47.9 (12.9)	48.6 (12.7)	.51
BMI	27.0 (4.2)	27.2 (4.2)	26.7 (4.0)	.28
Time in bed, min	479.9 (27.1)	479.9 (27.5)	479.9 (26.7)	.60
Sleep latency, min	16.1 (22.9)	16.7 (24.9)	15.1 (19.7)	.93
Sleep efficiency, min	79.9 (13.3)	79.5 (14.0)	80.4 (12.3)	.80
Apnea hypopnea index, events/h	32.6 (24.4)	32.9 (24.8)	32.0 (23.8)	.82
Apnea index, events/h	21.0 (21.8)	21.8 (22.9)	19.8 (20.2)	.57
Obstructive apnea index, events/h	18.3 (19.5)	18.8 (20.3)	17.4 (18.3)	.74
OSA severity, normal/mild/moderate/severe	43/80/109/191	26/46/72/110	17/34/37/81	.51

Abbreviations: BMI, body mass index, calculated as weight in kilograms divided by height in meters squared; OSA, obstructive sleep apnea.

Performance of Binary Classifiers for OSA

When the AHI threshold for binary classification of OSA was 5, 15, and 30 events/h, the accuracy of the OSA prediction was 88.2% ($\kappa = 0.46$), 82.3% ($\kappa = 0.59$), and 81.7% ($\kappa = 0.63$), respectively. Sensitivity was 90.8%, 87.3%, and 83.0%, and positive predictive value (PPV) was 95.8%, 89.3%, and 82.0% for AHI thresholds of 5, 15, and 30 events/h, respectively. Accuracy, sensitivity, PPV, F1 score, and area under precision recall curve tended to decrease as the threshold for OSA was increased. However, the κ value, specificity, and negative predictive value (NPV) increased along with the threshold: specificity was 64.7%, 70.6%, and 80.3%, respectively, for AHI thresholds of 5, 15, and 30 events/h, and NPV was 44.0%, 70.6%, and 81.3%, respectively. The AUC values were 0.90, 0.89, and 0.90, respectively, for AHI thresholds of 5, 15, and 30 events/h ([Table 2](#)).

Table 2. Performance of Binary Classification for Prediction of Obstructive Sleep Apnea From Sleep Breathing Sound Based on 3 Different Apnea Hypopnea Index Cutoff Values

Threshold	Accuracy, %	κ	Sensitivity, %	Specificity, %	PPV, %	NPV, %	F1 score	PRC area	AUC
5 Events/h	88.2	0.46	90.8	64.7	95.8	44.0	0.93	0.99	0.90
15 Events/h	82.3	0.59	87.3	70.6	89.3	70.6	0.87	0.95	0.89
30 Events/h	81.7	0.63	83.0	80.3	82.0	81.3	0.81	0.91	0.90

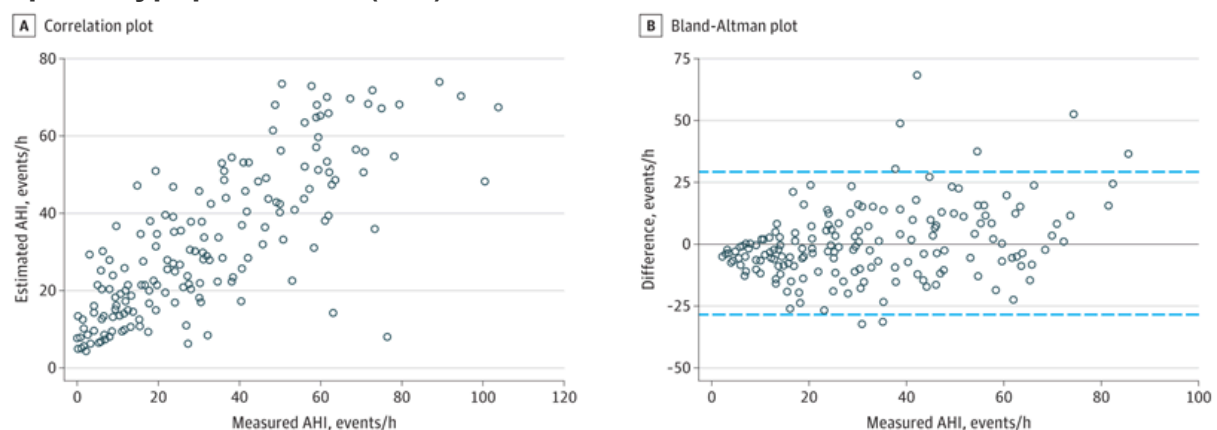
Abbreviations: AUC, area under curve; NPV, negative predictive value; PPV, positive predictive value; PRC area, area under precision recall curve.

AHI Estimation and Denoising and Attribute Selection

The AHI estimation was carried out using 4 regression models. Selected attributes (models 2 and 4) and their main characteristics are summarized in eTables 2 and 3 in the [Supplement](#).

All models had similar results, with a correlation coefficient ranging between 0.77 and 0.78. Using sleep breathing sounds without any denoising and attribute selection (model 4) yielded the best estimation performance. The correlation coefficient was 0.78, and the root mean squared error was 14.73 events/h. Other metrics are described in eTable 4 in the [Supplement](#). The correlation and Bland-Altman plots acquired from model 4 (feature selection without noise reduction) are described in [Figure 2](#). The Bland-Altman plot revealed that the mean difference between measured AHI and predicted AHI was approximately 0.23 (95% CI, -28.73 to 29.18) events/h, and error tended to increase as the mean AHI increased (coefficient, 0.24; 95% CI, 0.14-0.35; $R^2 = 0.11$), pointing to the underestimation of AHI as the OSA severity increases.

Figure 2. Correlation and Bland-Altman Plots of Estimated and Measured Apnea Hypopnea Index (AHI)



The estimation of AHI is based on model 4, which uses recorded sound without denoising followed by attribute selection. The correlation coefficient is 0.78 (A), and the error tended to increase as the mean AHI (mean of estimated AHI and measured AHI) increased (B) (coefficient, 0.24; 95% CI, 0.14-0.35; $R^2 = 0.11$). The small circles represent each case in the test data set. The reference lines on the Bland-Altman plot (B) show the mean (thin black line) and 95% limits of agreement (dashed blue lines).

The SHAP values calculated from model 4 are described in eFigure 1 in the [Supplement](#). The highest SHAP value of 15.45 was derived from the feature Derivative of Relative Difference Function Overall Standard Deviation.

Factors That Contributed to Error

A correlation analysis was carried out on several clinical parameters (AHI, sleep efficiency, BMI, and age), and error in AHI estimation (mean of the estimated AHI and measured AHI based on model 4). Factors that were statistically significantly associated with errors were AHI ($r = 0.59$; 95% CI, 0.48-0.72), sleep efficiency ($r = -0.17$; 95% CI, -0.32 to -0.02), BMI ($r = 0.36$; 95% CI, 0.22-0.50), and age ($r = 0.16$; 95% CI, 0.01-0.31) (eFigure 2 in the [Supplement](#)). A multivariate analysis using a linear regression model revealed that AHI ($\beta = 0.33$; 95% CI, 0.24-0.42) and sleep efficiency ($\beta = -0.20$; 95% CI, -0.35 to -0.05) were statistically significantly associated with error in estimation ([Table 3](#)).

Table 3. Multivariate Analysis Using a Linear Regression Model to Identify Factors Associated With Error

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Input variables	β (95% CI)
AHI ^a	0.33 (0.24 to 0.42)
Sleep efficiency	-0.20 (-0.35 to -0.05)
BMI	0.49 (-0.02 to 1.00)
Age	0.09 (-0.06 to 0.24)
Female sex	1.15 (-3.98 to 6.28)

Abbreviations: AHI, apnea hypopnea index; BMI, body mass index, calculated as weight in kilograms divided by height in meters squared.

^a The mean AHI is the mean of the estimated AHI and measured AHI.

Discussion

This study validated the use of breathing sounds recorded during sleep using a smartphone microphone for accuracy of OSA prediction models. To the best of our knowledge, this is the largest cohort of patients who simultaneously had their sleep

breathing sounds recorded by a smartphone during a standard full-night in-laboratory PSG.

Previous studies we have conducted included a small number of patients and sleep breathing sounds recorded on a low-quality microphone.^{8,9} We had conducted a 10-fold cross validation without validating through an independent test set that there would be a chance of overfitting. This study lowered the possibility of overfitting by separating the entire data set into training, validation, and test data sets.¹⁷ However, the patient distribution is somewhat different from that in the previous study.^{8,9} In the previous study, the number of patients was similar in each OSA severity group; however, in the current study, the number of patients tended to increase as the OSA severity increased. Imbalanced data could have affected accuracy. Instead, to overcome the imbalanced patient distribution, we used SMOTE in the training set to augment the learning from the minority class. The concept of SMOTE is the generation of synthetic data between each sample of the minority class and its k nearest neighbors. Therefore, the training data set was more balanced than before, and the problem of overfitting during model buildup was alleviated with SMOTE.¹⁴ Consequently, the OSA prediction performance was comparable with that in the previous study.^{8,9} The prediction accuracy of OSA ranged from 81.7% to 88.2%.

The presence of OSA with a cutoff value of 5 to 30 events/h could be predicted with a sensitivity of 83.0% to 90.8% and a specificity of 64.7% to 80.3% using breath sounds recorded with a smartphone during sleep. However, even though we balanced the data in the training set using SMOTE, the binary classification performance seemed to depend on the cutoff threshold. As the threshold was increased, sensitivity was lowered while specificity was increased. The typical trade-off between sensitivity and specificity was observed as the threshold value changed—decreasing sensitivity and increasing specificity as the threshold

increased. This has also been observed in other home sleep apnea test devices.¹⁸ In general, the prevalence of OSA is 28.4% in Korea and 33.2% in the US for AHI of 5 events/h or greater,¹⁹ which is lower than that of the current study. Therefore, predictive values would be adjusted when used for general population for screening purposes; PPV (82.0%-95.8%) would be further decreased as NPV (44.0%-81.3%) increased.

In the present study, several metrics, including accuracy, κ index, and receiver operation curves, were used to assess the predictive performance based on various AHI thresholds. Although accuracy, F1 score, area under precision recall curve, and AUC values were highest for AHI of 5 events/h or greater, the κ index, which measures the proportion of correct instances after accounting for the probability of chance agreement, was highest when the threshold for AHI was set up to 30 events/h or greater. In the previous study,⁸ where there had been an even distribution of the number of patients in each OSA severity category, the highest κ index was shown for AHI of 15 events/h or greater. The ratio between 2 classes above or below this level was 1:1, suggesting a balanced distribution for both training and testing. In the current study, although the input sound data was technically different, the majority of patients in the test data set were in the severe OSA category (47.9%). Therefore, when the cutoff value had been set at 30 events/h, the proportions of the classes below or above this level seemed to be the most balanced. Data distribution in the test data set influences the model performance²⁰ even after artificially balancing the data with SMOTE.

Overall, these results are comparable with portable home sleep test devices (sensitivity, 0.88; specificity, 0.88; AUC, 0.89 for AHI threshold of 15 events/h)²¹ and 1 recently published article about a wearable device based on reflective photoplethysmography (sensitivity, 0.73; specificity, 0.81; AUC, 0.86).²² Unlike

wearable or home sleep test devices, the present method is easier to implement and repeatable because breathing sounds can be easily obtained from a smartphone. Therefore, this method has potential for prescreening for OSA. However, even if the result is negative, for high-risk patients, such as adults with signs and symptoms (eg, excessive daytime sleepiness, obesity, habitual snoring, diagnosed hypertension) of OSA,²³ standard in-laboratory PSG or at least a home sleep apnea test should be recommended.

In the previous model based on a ceiling microphone in a PSG laboratory, input sound data underwent noise reduction followed by feature extraction.^{8,9} The process of noise reduction by spectral subtraction filtering and the extraction of all 508 sound features from sleep-recorded sound data demand additional resources. Because the ultimate goal was to predict or prescreen OSA using smartphones, it would be beneficial for the process to be simple. Interestingly, using sound data that had not been processed and had only selected features yielded a better prediction performance compared with that of sound data that had been denoised and used all 508 sound features. Model 4, which was based on sound data that had not been denoised and contained selected features, resulted in the best prediction performance. Attribute selection removes irrelevant features that actually decrease the prediction performance by introducing additional noise. This is known to decrease overfitting, improve accuracy, and reduce the training time.^{15,24} Most smartphones use adaptive noise cancellation by an additional microphone.²⁵ Adding a second microphone to sample the noise of the acoustic environment allows for that signal to be subtracted from the sound recorded using the original microphone.²⁶ Recorded sounds on smartphones have already been filtered once; therefore, noise reduction by spectral subtraction may not be necessary.

The highest SHAP from model 4 was derived from the overall standard deviation of the first derivative of the raw feature, named Relative Difference Function. This was also regarded as one of the features that could discriminate groups with severe OSA (AHI of ≥ 30 events/h) from others in the previous study.²⁷ Relative Difference Function is a feature that calculates the log of the derivatives of the root mean square. Because root mean square is used to calculate the amplitude of a signal, this feature is a useful tool for onset detection.²⁸ Because highest SHAP had been derived from the first derivative, temporal changes of this sound feature seem to be important.

This study also assessed the factors that are associated with error in AHI estimation based on model 4. In the Bland-Altman plot, the estimation error tended to increase as the mean of actual and predicted AHI increased, as in the previous study,^{8,9} which suggests the underestimation of AHI as the OSA severity increases. Although the largest proportion of the present cohort had severe OSA, the patient distribution according to AHI was skewed in both the training and test data sets. In contrast with binary classification, SMOTE had been performed for test data sets for regression. Patients whose AHI was very high (eg, >60 events/h) were less frequently identified. Thus, in the case of AHI, it appears that the biased patient distribution contributes to the overall data inconsistency, leading to estimation errors. In this case, further recruitment of patients with more severe OSA improves model performance.

Sleep efficiency was another factor associated with error in estimation, whereby patients with lower sleep efficiency tended to have a higher estimation error. For example, for a patient with considerably low sleep efficiency, the estimation model based on sleep breathing sounds would assume that the patient's sleep efficiency is within a normal range. However, when measured during routine PSG, AHI is calculated by the number of total respiratory events divided by total sleep time;

therefore, even if the sleep efficiency is low, AHI can be correctly measured. This would ultimately lead to a certain degree of underestimation of the AHI from sleep breathing sounds in patients with poor sleep efficiency. Analysis of sleep breath sounds using machine-learning technique was found to produce 87% concordance in macro sleep stages (rapid eye movement, non-rapid eye movement, and wake staging) compared with standard PSG.²⁹ Therefore, developing a model that can estimate both AHI and sleep stage simultaneously may reduce estimation errors.

Limitations

There were several limitations to this study. First, models were not generated according to specific clinical parameters such as sex. There are sex-specific differences in sleep breathing sounds.³⁰ One study³¹ developed models to predict the presence of OSA by analyzing snoring sounds, and the model performance was increased after considering the sex of patients. Because of the small number of female patients in the current study, we were unable to conduct a stratified analysis. In addition, sleep position may also affect acoustic features of sleep breathing sound,³² and this was not considered in the present study. Therefore, future studies should incorporate more female patients to assess the effects of sex and sleep positions. Second, the present model was based on sound recorded in a sleep laboratory, rather than the home environment, which is supposed to have a different noise background. Therefore, further validation should be completed in a real-life environment using various smartphone devices. Third, the smartphone microphone used in this study could not differentiate between sleep and waking state during recording. A time-specific analysis excluding the duration of the awake state in routine PSG may increase the overall accuracy. For this purpose, differentiation between sleep and awake state based on sleep breathing sounds is necessary. A recent study³³ showed that breathing movements can be recorded using a speaker and microphone within a mobile device by detecting changes in the echo of a

specially crafted phase-aligned modulated signal played via a selected speaker. This technology may be applied to differentiate sleep and awake states in the future.

The future direction of smartphone-based OSA prediction is to use it as a prescreening tool. Several evaluation metrics such as accuracy, negative and positive predictive values, and κ index may change as the disease prevalence changes. Therefore, the present model should be tested in a general population among patients of different races and ethnicities using a smartphone app. Another hurdle would be to minimize the calculation loading. One method would be to reduce the recording time. In the current study, we used 6 hours of recorded sound during sleep. Therefore, as in split-night PSG,³⁴ validating the first 2 to 3 hours of sleep breathing sound as an input would be reasonable.

Conclusions

In this cross-sectional study, we acquired a large data set of sleep breathing sounds recorded using a smartphone during full-night in-laboratory PSG and validated its usefulness in the prediction of OSA. These prediction models yielded a fair prediction performance, and we found that noise cancellation was not mandatory for a good prediction performance. The skewness of patient distribution in model development and inability to differentiate sleep and awake states should be addressed to improve the performance of the prediction model. Future research should be extended to incorporate real-life smartphone recordings at home using various smartphone devices.

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